Experimental and method

LOGICLAB KIDS — EOCYS 2026

Native Platforms

01

Native **SwiftUI (MVVM)** on iOS (live on Apple TestFlight) and **Kotlin Compose** on Android. Operates offline-first; all lessons execute with zero network connectivity.

Stage Grammar

02

6 modules and 36 lessons bundled as JSON. 4 interactive task types: Quiz, Spot-It, Sort-It, Build-It. Mistakes are never punished (explanations shown after 2 misses).

Premack Contingency

03

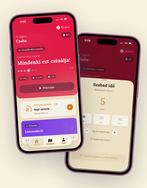
Applies the Premack (1959) differential probability law. Solving a lesson credits the ledger with **+20 play & +5 min**. Child spends balance within parent-allowed hours.

Constrained Gemini

04

System prompt restricts Gemini to Socratic dialogue: probes reasoning and gives hints, never direct answers. Strict low token cap prevents lecturing; unlocks only post-task.

Interactive Child Experience · iOS & Android



01 · DAILY MYSTERY

02 · EARN FREE TIME

The Behavioral Lock

Screen entertainment is the highest-probability behavior, structured reasoning practice is among the lowest. The lock and key are in the OS: a child cannot negotiate around it.

The Parent's Side

Governed, not policed

Daily limit
a hard ceiling on reward minutes

Allowed hours
when the minutes may be spent

App whitelist
which apps the shield lifts on

Per child
own login, own data only

FIG 3. The parent sets the economy once - daily ceiling, allowed window, whitelisted apps, one account per child. Everything after that is enforced by iOS FamilyControls, not by the app. Screens: shipped iOS build.

Opaque Token Shielding

The system picker returns **opaque tokens**, so LogicLab Kids never accesses which apps the parent selected. Shields are applied directly by iOS ManagedSettings, not in-app.

Inside a Lesson

Story → Task → Explanation

FIG 4. The "Slippery slope" lesson: the concept arrives inside a scene the child knows, becomes a task the child must solve, and a wrong answer ends in an explanation - never a punishment. The AI tutor names text, never first.

4 Core Task Types

Quiz (sound continuation), **Spot-It** (faulty step), **Sort-It** (inference chain), and **Build-It** (conclusion assembly). The child acts dynamically rather than memorizing definitions.

36	~5 min	100%	0 Net
Interactive Lessons	**Earned Screen-Time**	**OS-Level Shielding**	**Offline Runtime**
6 modules, across 3 levels	iOS and Android tested	Apple's FamilyControls & Gemini Tokens	Zero network required for lessons

REFERENCES

[1] Apple Inc. FamilyControls, ManagedSettings & DeviceActivity Frameworks.

[2] Premack, D. (1959). Toward empirical behavior laws I: Positive reinforcement.

[3] Kosmyna et al. (2025). Your Brain on ChatGPT. MIT Media Lab.

[4] LogicLab Kids System Architecture & Build Specification (2026).